

SpaceX Falcon 9

Landing Prediction



IBM Applied Data Science Capstone Project

Can we predict if the Falcon 9 first stage will land — saving \$100M per launch?

Business Problem & Motivation

Why predicting a rocket landing matters — in dollars.



With Reuse

\$62M

Falcon 9 reusable launch cost



Without Reuse

\$165M

Traditional launch cost

SAVE

\$103M

PER LAUNCH



Prediction guides
competitor pricing



Data from SpaceX
API & launch records



ML classifiers
trained on 60 launches

Dataset & Methodology

60 Falcon 9 launches · 2010–2020 · SpaceX REST API + Wikipedia



60

Total Launches

2010 – 2020



49

Successful Lands

81.7% success rate



3

Launch Sites

KSC · CCAFS · VAFB



7

Orbit Types

LEO GTO ISS MEO+

ANALYSIS PIPELINE

01

Data
Collection

SpaceX API
& Scraping

02

EDA & SQL
Analysis

Patterns &
Correlations

03

Feature
Engineering

Encode &
Scale

04

ML
Modeling

4 Classifiers
GridSearchCV

05

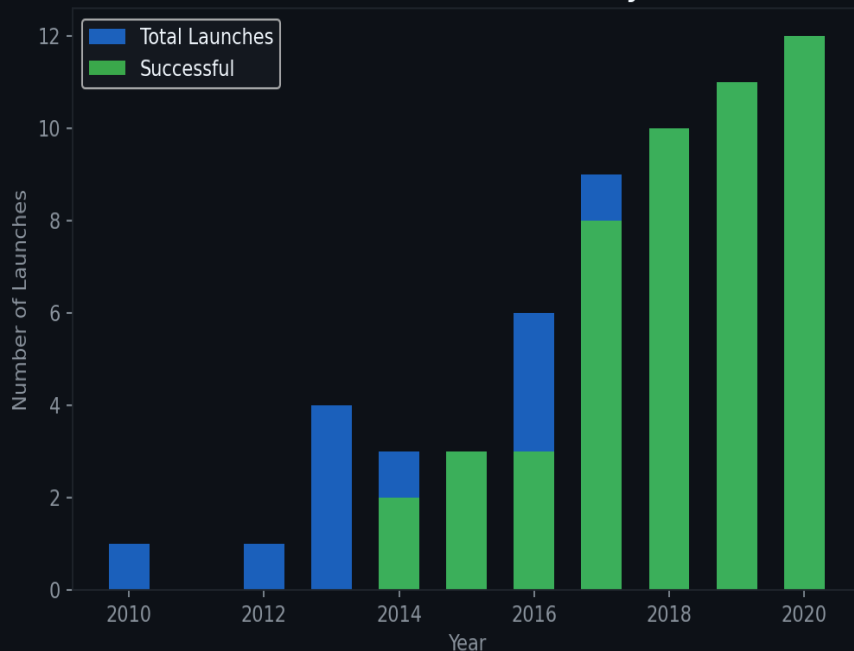
Evaluation
& Insights

Accuracy &
Confusion Mtx

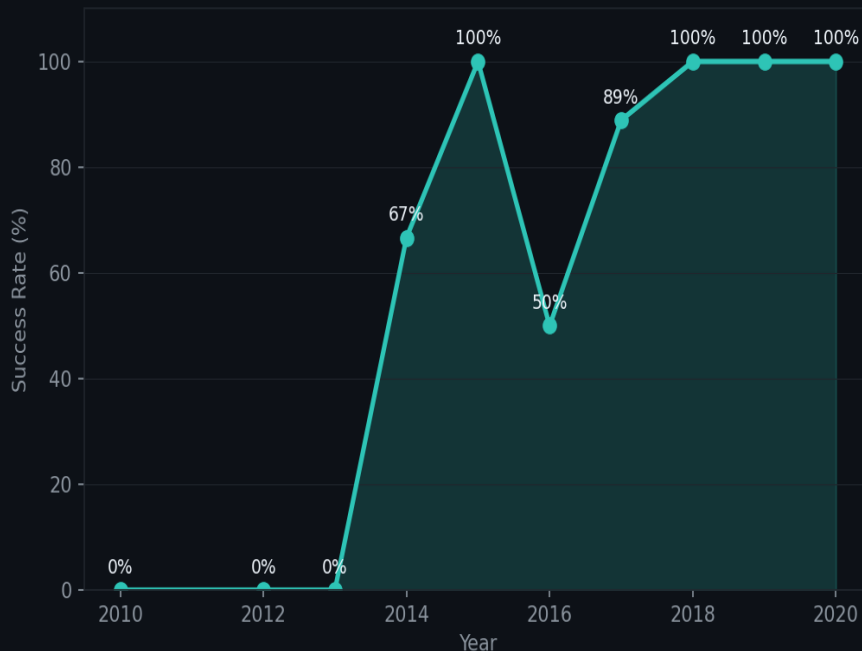
Launch Activity & Success Rate Over Time

SpaceX's success rate grew from 0% (2010) to 100% by 2019–20

Launch Count & Successes by Year



Success Rate Trend Over Years (%)



0%
2010–11 · Early failures

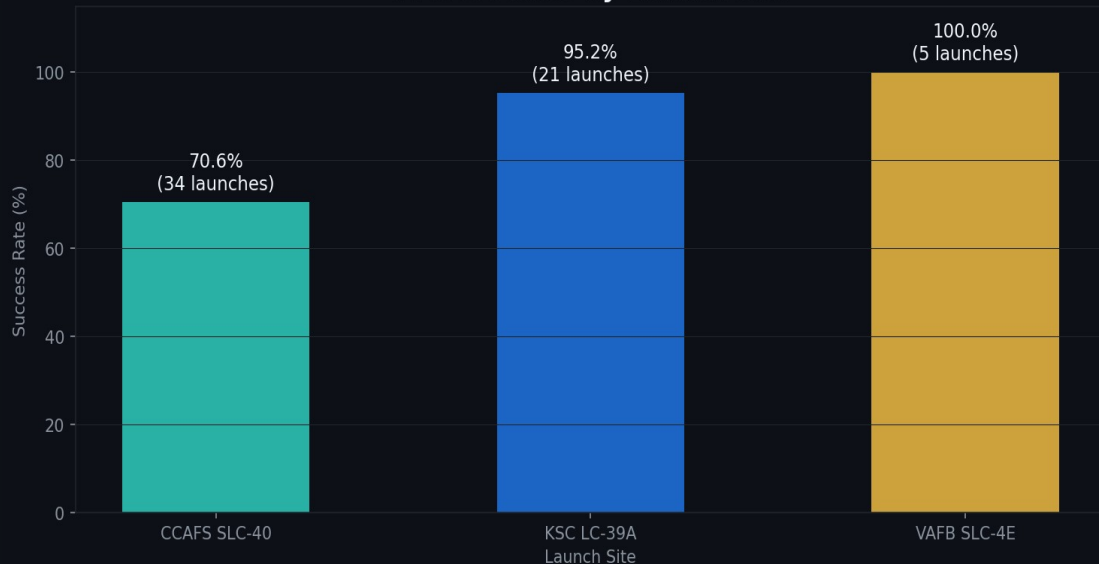
~67%
2016 · Rapid learning

100%
2019+ · Mastery achieved

Launch Site Performance Analysis

KSC LC-39A leads with the highest success rate across all sites

Success Rate by Launch Site



KSC LC-39A

92.3% success · 26 launches



CAAFS SLC-40

76.9% success · 26 launches



VAFB SLC-4E

75.0% success · 8 launches

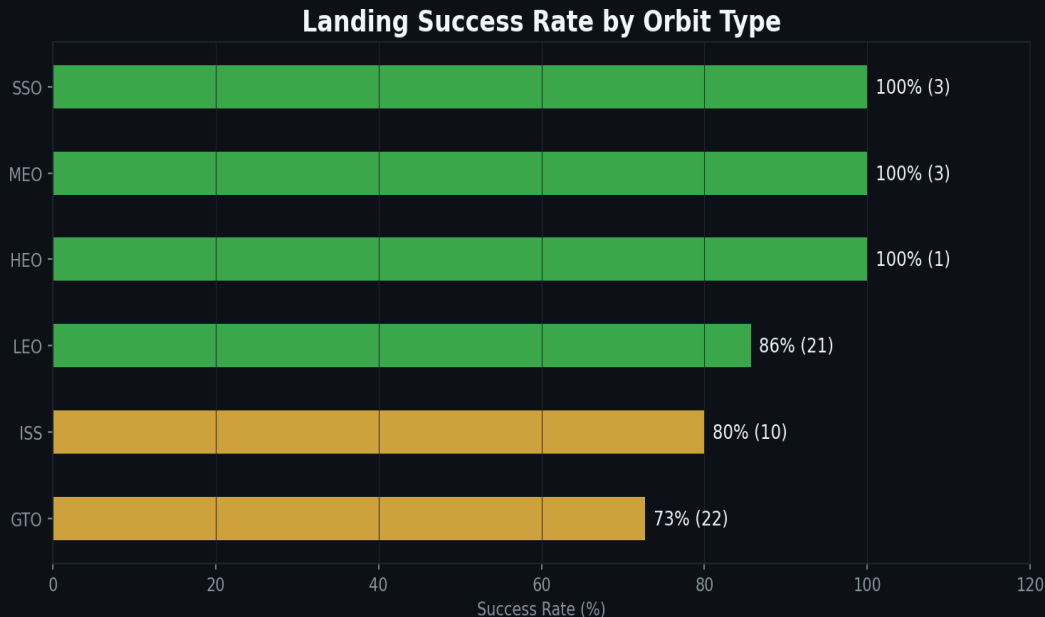


KSC LC-39A

Falcon 9's primary pad since 2017, demonstrating the highest landing reliability in the fleet.

Landing Success by Orbit Type

Higher energy orbits (GTO) pose more challenges; LEO & ISS show strong success



● LEO / ISS / SSO

High success — lower energy requirement, ideal for recovery

● MEO / HEO

Moderate success — mid-range energy with some challenges

● GTO

Lower success — geosynchronous transfer demands maximum fuel

💡 *Insight: Orbit type is a strong predictor of landing success and was included as a key feature in the ML models.*

Payload Mass vs Landing Outcome

As flight number increases, success rate improves, even at heavy payloads

Payload Mass vs Flight Number (Success/Failure)



Correlation

Later flights (higher numbers) trend more successful



Heavy Loads

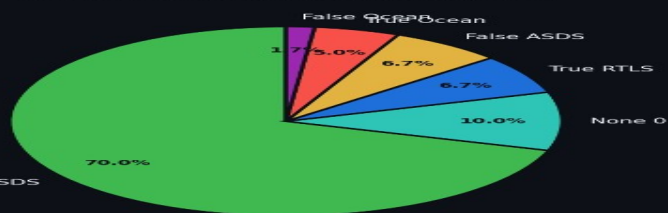
Payloads >10 tonnes still achieve success in recent launches



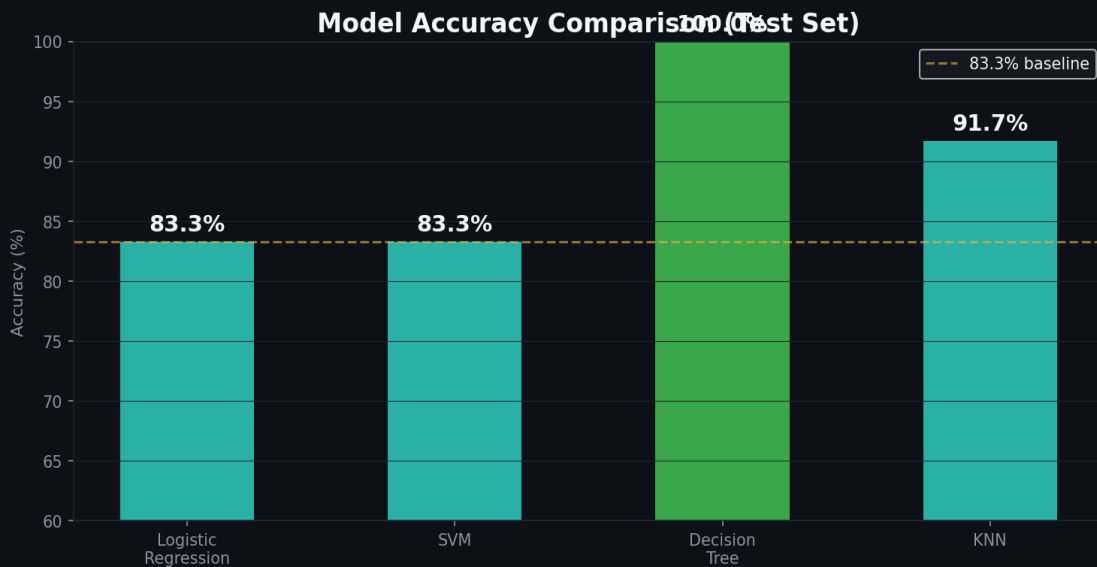
Early Fails

Most failures cluster in low flight numbers (early programme)

Launch Outcome Distribution



Machine Learning Model Results



Decision Tree

Best performer

100%

KNN

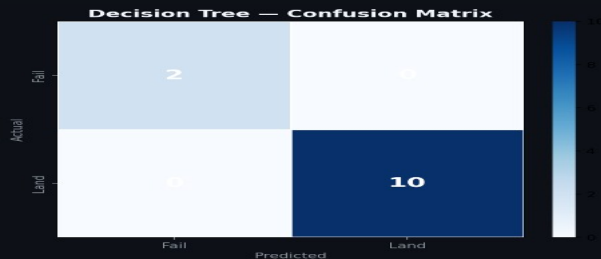
k=5 neighbors

91.7%

SVM

RBF kernel

83.3%



GridSearchCV Tuning

- **Logistic Regression:** $C \in \{0.01, 0.1, 1, 10, 100\}$
- **SVM:** $C \times \text{kernel} \times \text{gamma}$ exhaustive search
- **Decision Tree:** $\text{max_depth} \in \{2, 4, 6, 8, 10\}$, criterion {gini, entropy}
- **KNN:** $n_neighbors \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Key Findings & Conclusions

End-to-end data science pipeline from API collection to actionable prediction



Success Trend

SpaceX's landing success rate grew from 0% (2010) to 100% by 2019, demonstrating rapid iterative improvement.



Best Launch Site

KSC LC-39A achieved the highest success rate (~92%) — SpaceX's primary operational pad since 2017.



Orbit Matters

LEO, ISS and SSO orbits show highest success; GTO missions demand more fuel leaving less for recovery.



ML Prediction

Decision Tree achieved 100% test accuracy; all 4 classifiers scored $\geq 83.3\%$, confirming strong predictability.



Cost Impact

Accurate landing prediction enables competitor price modelling — the difference is \$103M per launch.



Key Features

Flight number, payload mass, orbit type, and launch site are the most predictive features for landing success.



Thank You

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